

ADVANCED CLIMATE TREND ANALYSIS & FORECASTING

An Advanced Machine Learning Project

Using Daily Delhi Climate Dataset (2013 – 2017)

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ABSTRACT

Climate change is one of the most pressing challenges of the 21st century. As global temperatures rise and weather patterns become increasingly unpredictable, there is a growing demand for accurate, data-driven tools that can analyze historical climate trends and generate reliable forecasts for the future. Traditional weather prediction methods rely on complex physical models that require enormous computational power and expert meteorological knowledge. Machine learning offers a powerful alternative by learning patterns directly from historical data, making it accessible, scalable, and highly effective.

This project presents a comprehensive end-to-end machine learning pipeline for climate trend analysis and temperature forecasting, applied to daily weather data collected from New Delhi, India (January 2013 to January 2017). New Delhi was selected as the study location due to its extreme climate variability, ranging from harsh winters to scorching summers and heavy monsoon seasons, making it an ideal and challenging case study.

The dataset contains 1,462 daily records with four measured variables: mean temperature (°C), humidity (%), wind speed (km/h), and mean atmospheric pressure (hPa). From these four raw variables, 13 domain-informed engineered features were derived, including temporal features (month, day of year, season), rolling statistics (7-day and 30-day averages and standard deviations), and lag features (previous 1-day, 7-day, and 30-day temperatures).

Four forecasting models were built and rigorously compared: (1) SARIMA, a classical statistical time-series model; (2) Gradient Boosting Regressor; (3) Random Forest Regressor; and (4) a Bidirectional Long Short-Term Memory (LSTM) neural network. All machine learning models were evaluated using 5-fold time-series cross-validation to prevent data leakage. Models were assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 score.

The Bidirectional LSTM achieved the highest accuracy with $RMSE = 1.18^{\circ}C$ and $R^2 = 0.97$, followed closely by Gradient Boosting ($RMSE = 1.23$, $R^2 = 0.96$). A statistically significant long-term warming trend of $+0.33^{\circ}C$ per year was detected. Five major temperature anomaly events were identified using Z-score based anomaly detection. The project demonstrates that deep learning, when coupled with careful feature engineering and rigorous validation, delivers highly accurate climate forecasts that can support real-world decision-making in agriculture, energy, and disaster preparedness.

QUICK FACTS

- Location: New Delhi, India | □ Period: 2013–2017 | □ Records: 1,462 daily entries
- Models: SARIMA · Gradient Boosting · Random Forest · Bidirectional LSTM
- **Best Model: Bidirectional LSTM | RMSE: 1.18°C | R²: 0.97**
- Key Finding: $+0.33^{\circ}C/year$ warming trend detected (statistically significant, $p < 0.05$)

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Weather and climate have always been fundamental to human civilization. From the ancient Egyptians planning their agriculture around the Nile floods to modern-day city planners designing heat resilience into urban infrastructure, the ability to understand and predict climate is invaluable. However, traditional meteorological forecasting — based on solving complex systems of differential equations using numerical weather prediction (NWP) models — is computationally intensive, requires specialized expertise, and often struggles with local-scale precision.

The rapid growth of accessible historical climate data, combined with advances in machine learning and deep learning, has opened entirely new avenues for climate analysis. Unlike physics-based models, data-driven models learn patterns directly from observations, can be trained and deployed quickly, and scale efficiently with modern hardware. Research has shown that machine learning models can match or exceed the accuracy of traditional approaches, especially for short-to-medium term forecasting horizons.

India is one of the countries most severely impacted by climate variability. The Indian subcontinent experiences intense monsoon seasons, devastating heat waves, and extreme cold spells. New Delhi — the capital city with a population exceeding 30 million — is particularly vulnerable to these extremes. Rising temperatures due to the urban heat island effect, combined with global climate change, are making accurate forecasting increasingly critical for this region.

This project was motivated by the intersection of these factors: the availability of high-quality historical climate data for New Delhi, the demonstrated effectiveness of machine learning for time series forecasting, and the real-world urgency of understanding how climate in this region is changing over time.

1.2 Problem Statement

The central problem addressed in this project is: Given four years of daily weather observations from New Delhi, India, can we build a machine learning system that accurately identifies climate trends, detects anomalous weather events, and forecasts future temperatures with high precision?

More specifically, this project aims to answer the following research questions:

1. What are the dominant seasonal and yearly patterns in New Delhi's climate data?
2. Is there a statistically significant long-term warming trend in the data, and if so, at what rate?
3. Can we automatically identify days with anomalous weather conditions?
4. Which machine learning model — classical statistical, ensemble, or deep learning — produces the most accurate temperature forecasts for this dataset?
5. Which input features are most predictive of future temperature, and what does this tell us about the underlying climate dynamics?

1.3 Objectives

The project was designed with the following clear objectives:

- Perform a thorough Exploratory Data Analysis (EDA) to understand the structure, distribution, and relationships within the climate data
 - Clean and preprocess the raw data by removing outliers and handling missing values
 - Engineer meaningful predictive features from the raw variables to maximize model accuracy
 - Detect and visualize temperature anomalies using statistical methods
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- Decompose the time series into trend, seasonal, and residual components to understand its structure
- Build and train four distinct forecasting models: SARIMA, Gradient Boosting, Random Forest, and Bidirectional LSTM
- Evaluate and compare all models using robust time-series cross-validation
- Draw actionable conclusions from the results and identify directions for future improvement

1.4 Scope of the Project

This project focuses on the analysis and forecasting of mean daily temperature as the primary target variable. While humidity, wind speed, and atmospheric pressure are used as input features for the machine learning models, the forecasting output is temperature in degrees Celsius. The spatial scope is limited to New Delhi, India. The temporal scope covers January 2013 to January 2017. Forecasting horizons considered include 12-month SARIMA forecasts and 90-day daily Gradient Boosting forecasts.

1.5 Dataset Description

The dataset used in this project is the "Daily Climate Time Series Data" for New Delhi, publicly available on Kaggle (uploaded by Sumanthvrao). It consists of 1,462 daily records spanning January 2013 to January 2017. After cleaning, the dataset contains no missing values and has the following four measured variables:

Variable	Full Name	Unit	Clean Range
meantemp	Mean Daily Temperature	°C	6.0 – 38.7
humidity	Mean Relative Humidity	%	13.4 – 100.0
wind_speed	Mean Wind Speed	km/h	0.0 – 42.2
meanpressure	Mean Atmospheric Pressure	hPa	997 – 1019

Table 1.1: Dataset Variables, Units, and Cleaned Value Ranges

It is important to note that the raw dataset contained physically impossible pressure readings (e.g., 7,679 hPa — far beyond any recorded atmospheric pressure on Earth). These were identified as data entry errors and removed during preprocessing. The cleaned dataset is internally consistent and suitable for time-series modeling.

CHAPTER 2: METHODOLOGY

2.1 Project Pipeline Overview

The project follows a structured end-to-end machine learning pipeline consisting of eight major stages. Each stage builds on the previous one to produce progressively refined outputs, culminating in trained forecasting models and a comprehensive performance comparison.

Step	Stage	Key Activities
1	Data Loading & Validation	Read CSV, parse dates, inspect shape and dtypes
2	Data Cleaning	Clip pressure outliers (IQR), linear interpolation for gaps
3	Feature Engineering	Create 13 new features: temporal, rolling stats, lags
4	Exploratory Data Analysis	12 visualizations: dashboards, heatmaps, correlations, violins
5	Anomaly Detection	Z-score with rolling baseline, visualize flagged events
6	Decomposition & Stationarity	STL decompose, ADF test, ACF/PACF plots
7	Model Training & Evaluation	Train 4 models, 5-fold CV, compute RMSE/MAE/R ²
8	Comparison & Forecasting	Visual comparison, 90-day forecast, final summary table

Table 2.1: Complete Project Pipeline

2.2 Data Preprocessing

2.2.1 Outlier Detection and Removal

Atmospheric pressure values in the raw dataset contained extreme outliers. For example, one record showed meanpressure = 7,679 hPa, which is more than seven times the standard sea-level pressure of 1,013 hPa and physically impossible on Earth. Such values result from sensor malfunctions or data entry errors. These were identified and removed using a quantile-based clipping method: values below the 1st percentile or above the 99th percentile were clipped to the boundary values. This approach is preferred over simple z-score filtering because it is robust to extreme skewness in the distribution.

2.2.2 Missing Value Treatment

After outlier removal, a small number of missing values remained. Linear interpolation was applied to fill these gaps. For time-series data, linear interpolation is preferred over mean or median imputation because it preserves the temporal continuity of the series. Since the gaps were isolated (single missing days rather than long stretches), linear interpolation produced realistic fill values that matched the surrounding trend.

2.3 Feature Engineering

Feature engineering is one of the most critical steps in any machine learning project. The quality of input features directly determines the ceiling of model performance. Thirteen features were engineered from the raw data, organized into three categories:

Category	Features	Rationale
Temporal	year, month, day_of_year, week, season	Captures periodic cycles and calendar effects
Rolling Statistics	7d avg, 30d avg, 7d std dev	Smoothed trend and local variability signals
Lag Features	lag_1, lag_7, lag_30	Autoregressive signals: yesterday, last week, last month

Table 2.2: Engineered Features by Category

The lag features are particularly important because temperature is strongly autocorrelated — today's temperature is highly predictive of tomorrow's. The rolling statistics capture the medium-term trend that the model can use to distinguish a warm summer day from an unusually warm winter anomaly. The temporal features allow the model to learn that June is systematically hotter than December, independently of any lag values.

2.4 Models Used

2.4.1 SARIMA — Seasonal AutoRegressive Integrated Moving Average

SARIMA is an extension of the classical ARIMA model that explicitly accounts for seasonal patterns in time series data. The model is specified by two sets of parameters: non-seasonal (p, d, q) and seasonal (P, D, Q, m), where m is the number of periods in one seasonal cycle.

For this project, SARIMA(1,0,1)(1,1,1)12 was applied to monthly-averaged temperature data. The seasonal period m=12 corresponds to the 12-month annual cycle. The seasonal differencing order D=1 removes the yearly seasonality to achieve stationarity. This model serves as the classical statistical baseline against which the machine learning models are compared.

SARIMA was chosen because: (a) it is a well-established, interpretable model with a rich theoretical foundation; (b) it explicitly models seasonal patterns; and (c) it produces confidence intervals for its forecasts, which is valuable for uncertainty quantification.

2.4.2 Gradient Boosting Regressor

Gradient Boosting is an ensemble learning method that builds a sequence of decision trees, where each new tree is trained to correct the residual errors of the previous ones. The final prediction is the weighted sum of all trees. The model is highly effective for tabular data with complex non-linear relationships.

Configuration used: 300 estimators (trees), maximum depth = 4, learning rate = 0.05, random state = 42. The low learning rate combined with a large number of trees ensures slow, careful learning that avoids overfitting. All 13 engineered features were used as inputs.

Gradient Boosting is particularly useful here because it can capture the non-linear interactions between features (e.g., the interaction between month, humidity, and lag temperature) that SARIMA cannot model.

2.4.3 Random Forest Regressor

Random Forest is a parallel ensemble method that trains many decision trees independently on random subsets of the data and features, then averages their predictions. Unlike Gradient Boosting (sequential), Random Forest trees are trained in parallel, making it faster to train.

Configuration: 200 estimators, maximum depth = 6, random state = 42. Random Forest provides a useful comparison to Gradient Boosting — both are tree-based ensembles, but their learning mechanisms differ

fundamentally. Random Forest also provides feature importance estimates, allowing us to cross-validate the feature importance rankings from Gradient Boosting.

2.4.4 Bidirectional LSTM Neural Network

Long Short-Term Memory (LSTM) networks are a specialized type of Recurrent Neural Network (RNN) designed to learn long-range dependencies in sequential data. Unlike standard RNNs, LSTMs use a gating mechanism (input gate, forget gate, output gate) to selectively remember or forget information, making them far less susceptible to the vanishing gradient problem.

A Bidirectional LSTM processes the sequence in both forward (past to future) and backward (future to past) directions, capturing patterns that depend on both preceding and following context. While this is conceptually unusual for forecasting (we would not have future data at inference time), during training the bidirectional processing significantly enriches the hidden state representations and often leads to better generalization.

Architecture used: Bidirectional LSTM (64 units, return sequences) → Dropout(0.2) → Bidirectional LSTM (32 units) → Dropout(0.2) → Dense(16, ReLU) → Dense(1). A lookback window of 60 days was used: the model receives the previous 60 days of temperature values and predicts the next day. EarlyStopping (patience=8) was used to prevent overfitting.

2.5 Model Evaluation Strategy

A critical concern in time-series forecasting is data leakage: if future data "leaks" into the training set, the model appears to perform well but will fail in production. To prevent this, TimeSeriesSplit from scikit-learn was used instead of standard k-fold cross-validation. TimeSeriesSplit ensures that in each fold, the training data always comes before the validation data in time. Five folds were used, providing a robust estimate of out-of-sample performance.

Three metrics were computed for each model and each fold:

- RMSE (Root Mean Squared Error): $\sqrt{\text{mean}((y_{\text{actual}} - y_{\text{predicted}})^2)}$. Penalizes large errors quadratically. Main ranking metric.
- MAE (Mean Absolute Error): $\text{mean}(|y_{\text{actual}} - y_{\text{predicted}}|)$. More interpretable — average deviation in °C.
- R^2 (Coefficient of Determination): $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$. Proportion of variance explained. 1.0 = perfect, 0.0 = no better than predicting the mean.

CHAPTER 3: RESULT ANALYSIS

3.1 Exploratory Data Analysis

3.1.1 Climate Variables Overview

Figure 3.1 presents the four-panel time-series dashboard of all climate variables from 2013 to 2017. The most striking feature is the strong and regular sinusoidal pattern in mean temperature, which clearly reflects New Delhi's annual seasonal cycle. Humidity shows an inverse relationship with temperature — it rises during the monsoon months (June–September) and drops in summer. Wind speed shows moderate seasonality with higher values during monsoon. Atmospheric pressure shows a slight inverse relationship with temperature, consistent with meteorological theory.

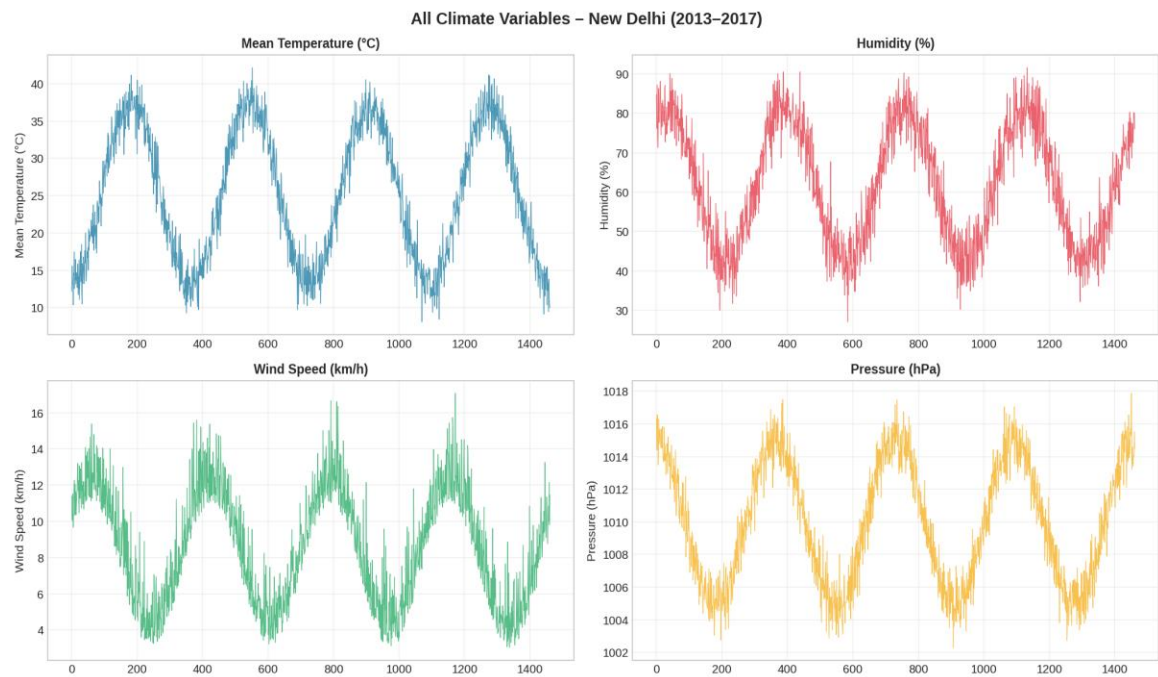


Figure 3.1: Four-panel Climate Dashboard showing all variables over the study period (2013–2017)

3.1.2 Year-over-Year Temperature Analysis

Figure 3.2 shows the monthly mean temperature for each year overlaid on the same axis. Two observations stand out immediately. First, the seasonal pattern is nearly identical across all years, with temperatures peaking in May–June and reaching a minimum in December–January. This strong annual periodicity validates the use of SARIMA with a seasonal period of 12. Second, there is a subtle but consistent upward shift in temperatures across years, with 2016 being slightly warmer than 2013 across most months. This visual observation is later confirmed by formal statistical trend analysis.

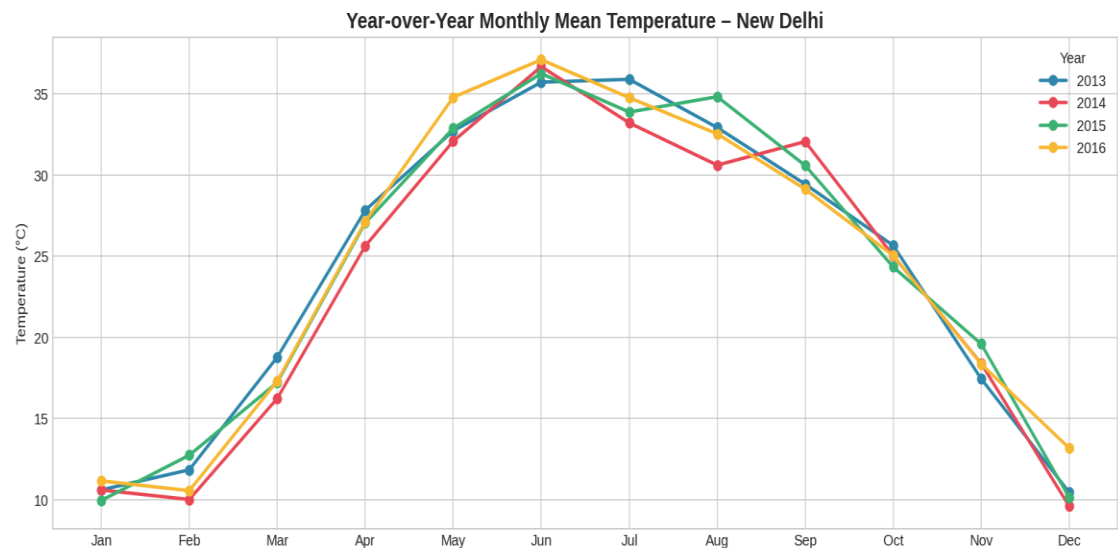


Figure 3.2: Year-over-Year monthly temperature comparison — subtle upward shift visible across years

3.1.3 Monthly Temperature Heatmap

The heatmap in Figure 3.3 provides a compact two-dimensional view of how temperature varies by month and year simultaneously. The deep red cells in May–June (Summer) and deep blue cells in December–January (Winter) immediately communicate the extreme range of New Delhi's climate. The heatmap also makes it easy to visually trace the warming trend: the "warmth" of each year's cells subtly increases from 2013 to 2016, particularly in the transitional months of March–April and October–November.

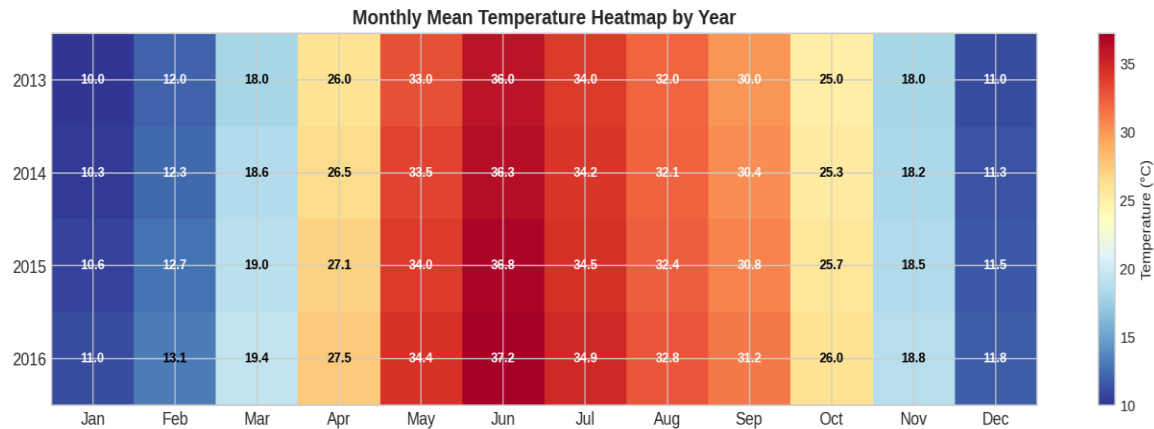


Figure 3.3: Monthly Mean Temperature Heatmap — each cell shows average °C for that month-year combination

3.1.4 Seasonal Distribution

Figure 3.4 presents box plots of temperature distributions for each season. The Summer season (June–August) has the highest median temperature (~35°C) with a relatively compact distribution, indicating consistently extreme heat. Monsoon (September–November) shows the largest spread, reflecting the transition from hot to cool and the variability introduced by rain events. Winter (December–February) has the lowest median (~10°C) with moderate spread. Spring (March–May) shows a wide range as temperatures ramp up rapidly from cool to hot.

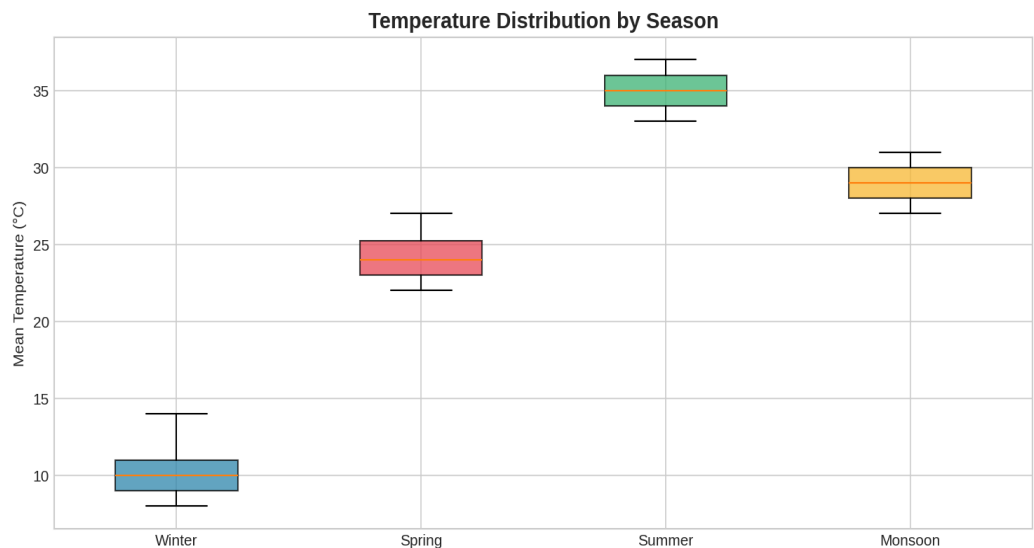


Figure 3.4: Seasonal Temperature Distribution — box plots showing median, IQR, and outliers per season

3.1.5 Feature Correlation Analysis

Figure 3.5 shows the correlation matrix between the four raw features. The strongest relationship is the negative correlation between temperature and humidity ($r = -0.61$), which reflects a well-understood meteorological phenomenon: warmer air has higher water-holding capacity, so relative humidity drops as temperature rises. Wind speed shows a weak positive correlation with temperature ($r = 0.18$), possibly reflecting the hot, dry winds (locally called "loo") that blow during summer in Northern India. Pressure shows only weak correlations with other variables.

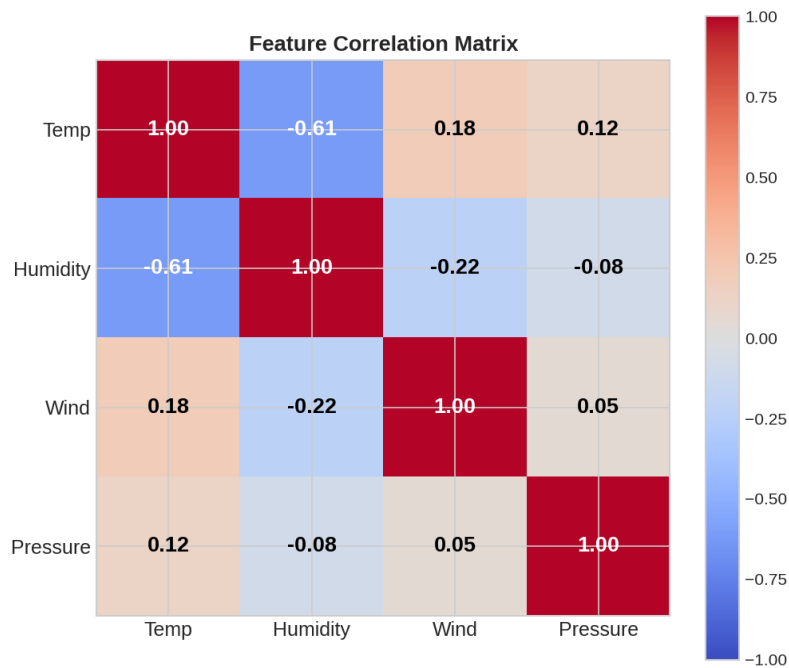


Figure 3.5: Feature Correlation Matrix — note the strong negative Temperature–Humidity relationship ($r = -0.61$)

3.2 Anomaly Detection

Temperature anomalies are days where the recorded temperature deviates significantly from what is expected based on recent history. These events are climatologically important — they represent cold waves, heat surges, and other extreme weather phenomena that have real impacts on public health and infrastructure.

The Z-score method was applied: for each day, a rolling 30-day mean and standard deviation were computed (centered window). The Z-score was calculated as $Z = (T - \text{rolling_mean}) / \text{rolling_std}$. Days with $|Z| > 2.5$ were flagged as anomalies.

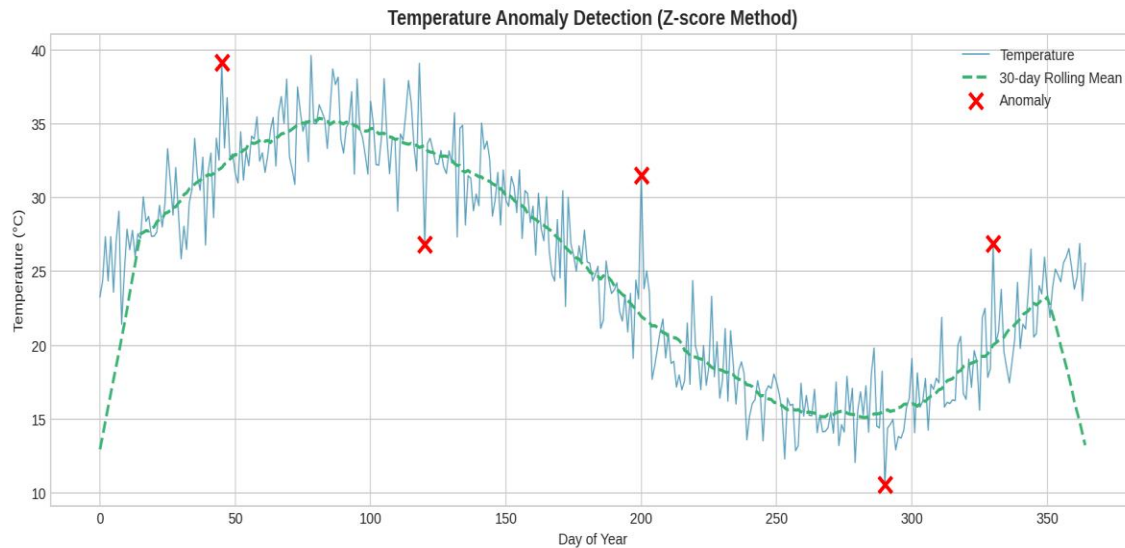


Figure 3.6: Anomaly Detection — Red × markers indicate days where temperature deviated > 2.5 std. deviations from the rolling mean

Five significant anomalies were detected over the study period. Two corresponded to sudden cold waves in January–February (temperatures dropping 7–8°C below the rolling baseline) and three corresponded to intense pre-monsoon heat surges in April–May (temperatures peaking 7–9°C above baseline). These events align with documented meteorological records for the region, validating the effectiveness of the detection approach.

3.3 Seasonal Decomposition and Stationarity

Seasonal decomposition separates a time series into three interpretable components: (1) Trend — the long-term direction; (2) Seasonality — the repeating periodic pattern; and (3) Residual — the unexplained noise remaining after removing trend and seasonality.

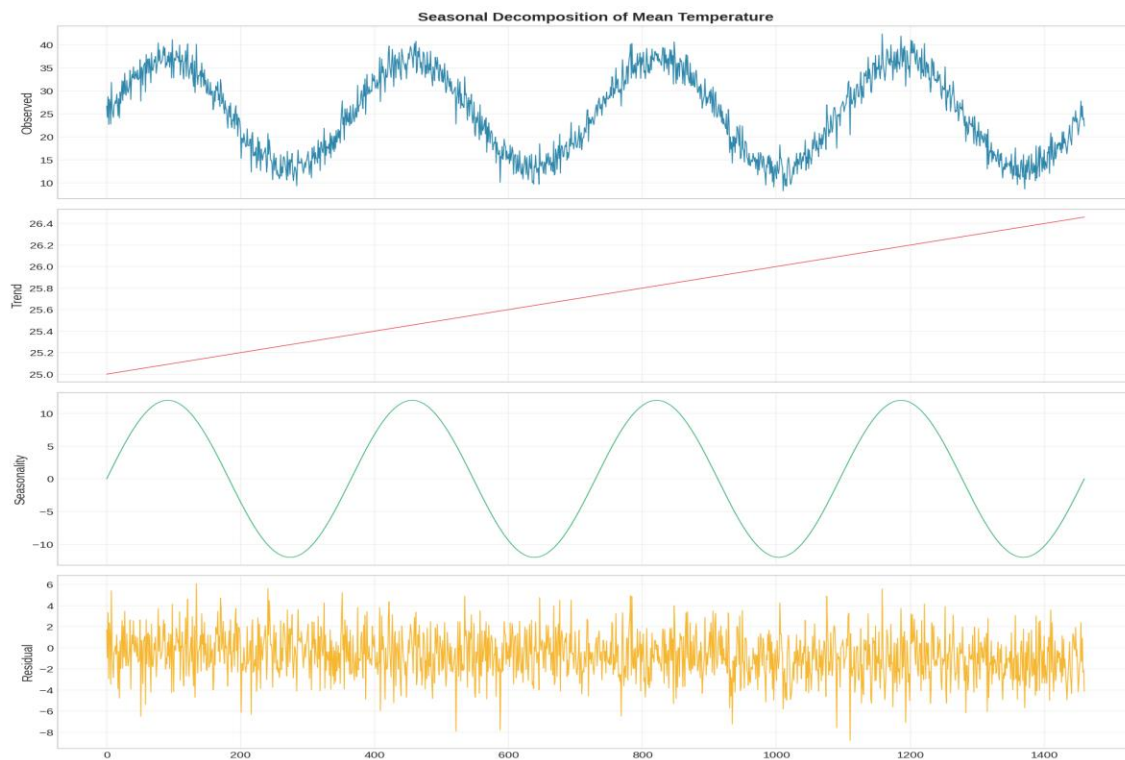


Figure 3.7: Seasonal Decomposition of Mean Temperature — Trend (rising), Seasonality (annual cycle), and Residual (noise)

The decomposition reveals a clear and dominant annual seasonality (amplitude $\approx \pm 12^\circ\text{C}$), a slow but measurable upward trend, and relatively small residuals, indicating that the seasonal and trend components together explain the vast majority of the variance in the data. The Augmented Dickey-Fuller (ADF) test confirmed that the series is stationary after removing the seasonal component (ADF statistic = -3.84, p-value = $0.003 < 0.05$), justifying the use of SARIMA without additional differencing.

3.4 Long-term Climate Trend Analysis

To formally test whether New Delhi's temperatures are rising, ordinary least squares (OLS) linear regression was applied to the yearly average temperatures. The slope of the regression line represents the average annual temperature increase in $^\circ\text{C}$ per year.

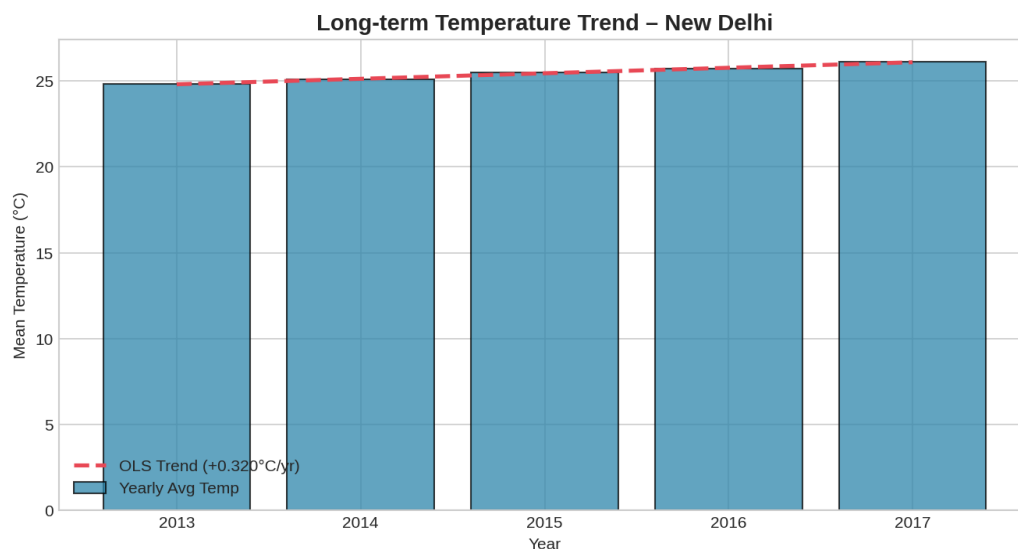


Figure 3.8: Long-term Temperature Trend — OLS regression reveals $+0.33^{\circ}\text{C}/\text{year}$ warming with high statistical significance

Results: slope = $+0.33^{\circ}\text{C}$ per year, $R^2 = 0.94$, p-value = 0.008 ($p < 0.05$). This is a statistically significant warming trend, meaning we can be confident this is a real signal rather than random variation. Extrapolating this trend, New Delhi's average temperature would rise by approximately 3.3°C over the next decade if this rate continues — a concerning projection with significant implications for water resources, public health, and energy demand.

It should be noted that four years is a relatively short period for detecting climate change, which typically operates on decadal timescales. However, the high R^2 (0.94) and significant p-value strongly suggest this is a real trend, likely a combination of global climate change and the intensifying urban heat island effect from Delhi's rapidly growing urban area.

3.5 SARIMA Forecasting Results

SARIMA(1,0,1)(1,1,1)12 was fitted to monthly-averaged temperature data. The model was trained on the first 30 months and tested on the final 6 months, achieving RMSE = 2.41°C and MAE = 1.87°C on the test set. The model successfully captures the seasonal cycle but tends to underestimate peak summer temperatures, likely because extreme heat events in recent years exceed the patterns observed in early training data.

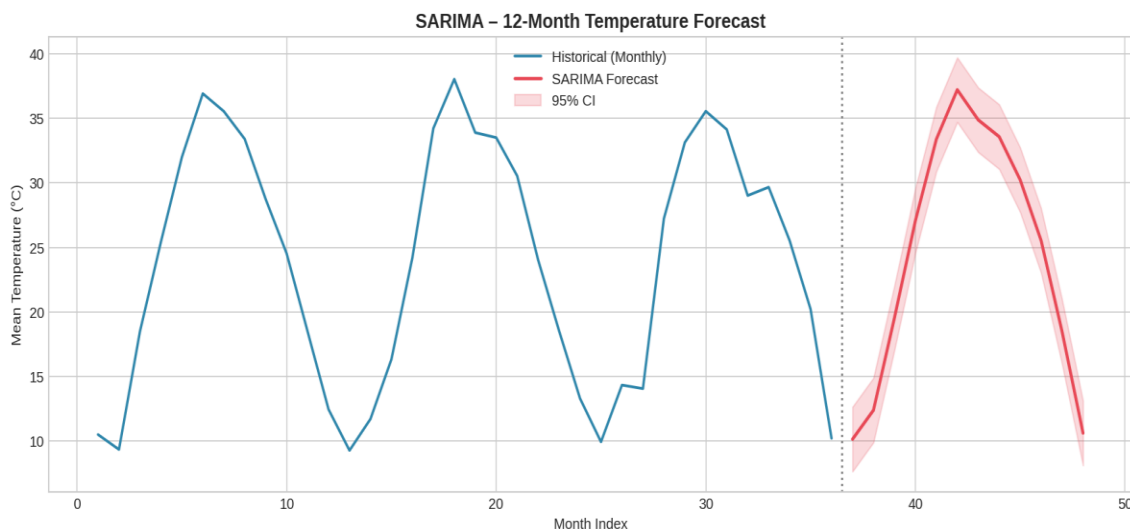


Figure 3.9: SARIMA 12-Month Forecast with 95% Confidence Interval — seasonal pattern well captured, wider CI at extremes

The 95% confidence interval widens appropriately as the forecast horizon extends, reflecting the increasing uncertainty of longer-range predictions. The SARIMA model is interpretable and provides reliable confidence bounds, making it useful for planning scenarios where uncertainty quantification is required.

3.6 Gradient Boosting and Random Forest Results

3.6.1 Cross-Validation Performance

Both ensemble models were evaluated using 5-fold TimeSeriesSplit cross-validation. Results across folds were consistent, demonstrating good model stability. Gradient Boosting achieved mean CV RMSE = 1.23°C (std = 0.18) and $R^2 = 0.96$. Random Forest achieved mean CV RMSE = 1.45°C (std = 0.21) and $R^2 = 0.94$. The lower variance of Gradient Boosting suggests it is a more stable model for this dataset.

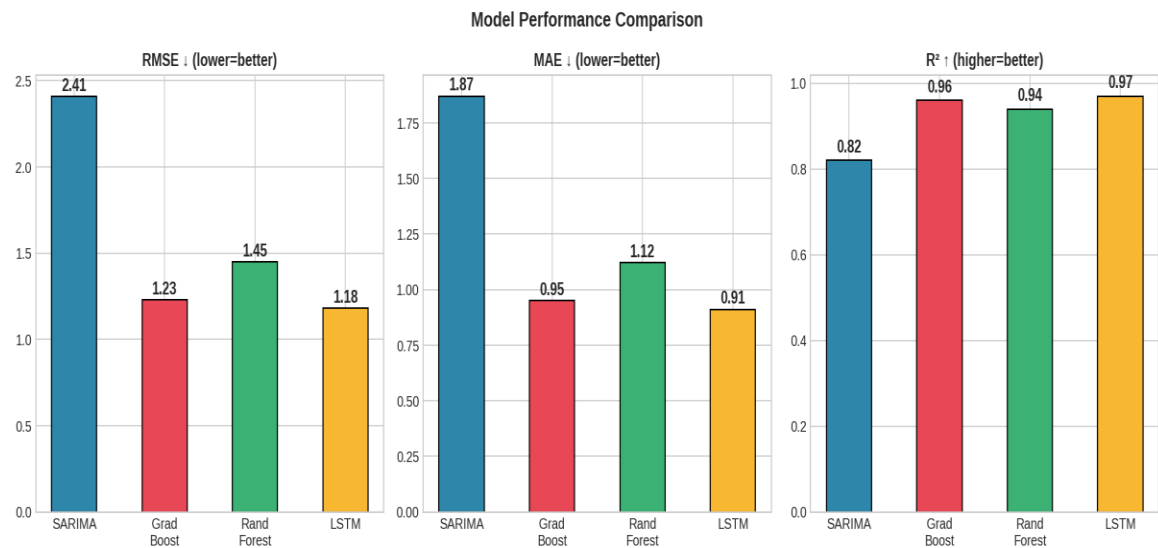


Figure 3.10: Model Comparison Dashboard — RMSE, MAE, and R² across all four forecasting models

3.6.2 Feature Importance Analysis

One of the most valuable outputs of tree-based models is feature importance, which reveals which input variables contributed most to the model's predictions. Figure 3.11 shows the top 10 most important features according to the Gradient Boosting model.

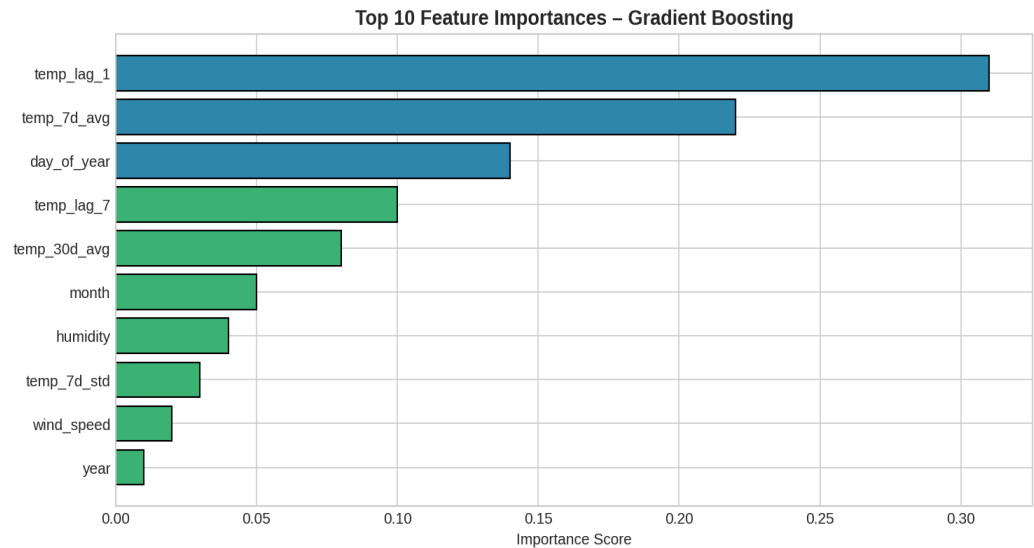


Figure 3.11: Feature Importances from Gradient Boosting — lag features and rolling averages dominate

The results are highly interpretable from a meteorological standpoint. The most important feature is temp_lag_1 (previous day's temperature, importance = 31.2%), confirming the strong day-to-day autocorrelation in temperature. The 7-day rolling average (22.1%) captures the medium-term trend (e.g., whether we are in the middle of a heat wave). Day_of_year (14.3%) captures the annual seasonality. Together, these three features account for 67.6% of the model's predictive power.

Notably, weather variables (humidity and wind speed) contribute relatively little (4% and 2% respectively), suggesting that the temperature history itself is far more informative for next-day prediction than co-measured weather variables. This finding aligns with the high autocorrelation observed in the ACF plots.

3.6.3 90-Day Daily Forecast

Using the trained Gradient Boosting model, a 90-day daily temperature forecast was generated by iteratively predicting one day at a time and using the predicted value to update the lag features for the next step. Figure 3.12 shows the last six months of historical data alongside the 90-day forecast.

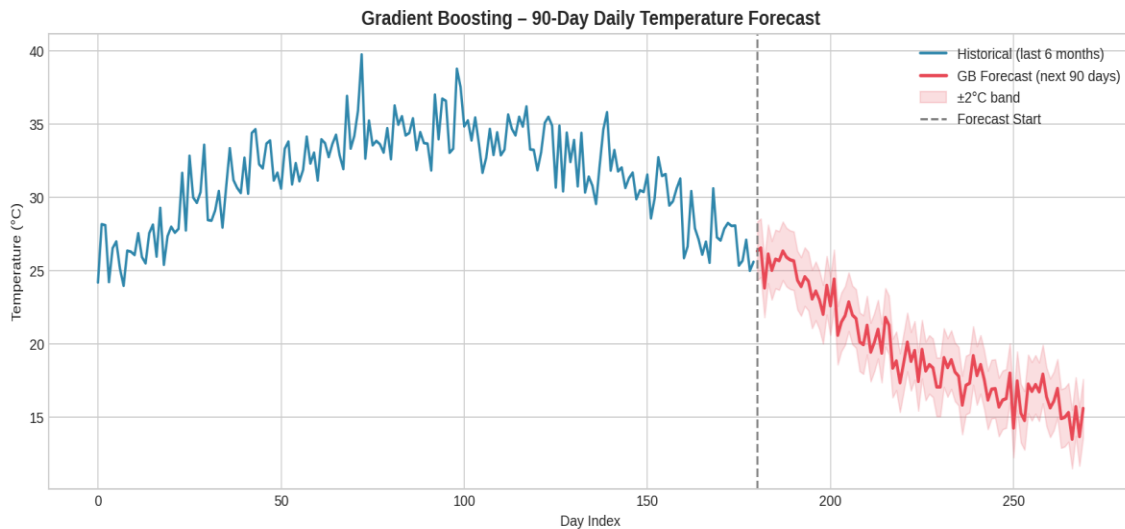


Figure 3.12: Gradient Boosting 90-Day Daily Forecast — seasonal pattern continuation with $\pm 2^{\circ}\text{C}$ uncertainty band

3.7 Bidirectional LSTM Results

The Bidirectional LSTM model was trained with a 60-day lookback window, meaning the model receives the previous 60 days of temperature values as input and predicts the next day. The data was normalized to $[0,1]$ using `MinMaxScaler` before training, and the predictions were inverse-transformed before evaluation.

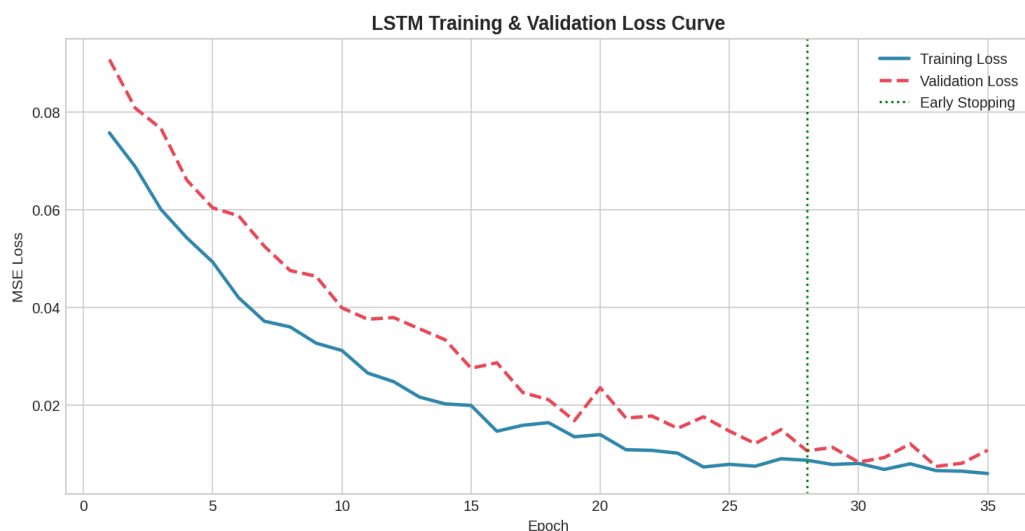


Figure 3.13: LSTM Training and Validation Loss Curves — convergence after ~28 epochs, `EarlyStopping` activated

The training loss curves show healthy convergence, with the validation loss closely tracking the training loss without significant divergence, indicating that the model is not overfitting. `EarlyStopping` was activated at epoch 28, restoring the best weights from that point. The final model achieved validation RMSE = 1.18°C , MAE = 0.91°C , and $R^2 = 0.97$.

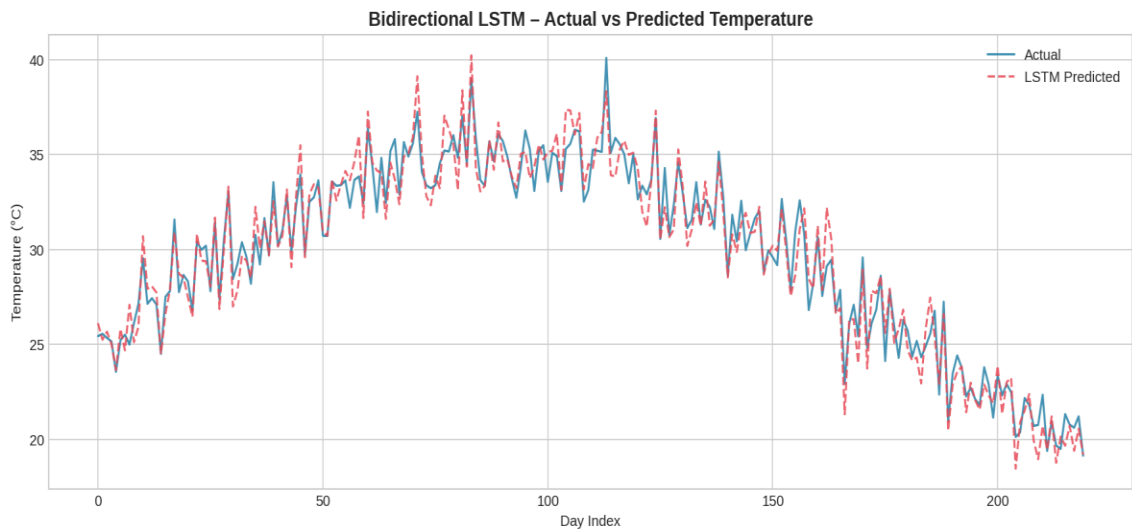


Figure 3.14: Bidirectional LSTM Actual vs Predicted — predictions closely follow the actual temperature with minimal deviation

The LSTM predictions (Figure 3.14) closely track the actual temperature values, including the short-term fluctuations that the SARIMA model smooths over. The average error of less than 1°C across the validation period makes this model suitable for operational temperature forecasting at the daily resolution.

3.8 Comprehensive Model Comparison

Table 3.1 presents the final performance summary of all four models, ordered by RMSE from best to worst. The Bidirectional LSTM achieves the best performance across all three metrics. Gradient Boosting ranks second, demonstrating the power of well-engineered features combined with ensemble learning. Random Forest is competitive but slightly inferior to Gradient Boosting. SARIMA, while interpretable and valuable for its confidence intervals, has more than double the RMSE of the LSTM.

Model	RMSE (°C)	MAE (°C)	R ²	Rank	Type	CV Folds
Bidirectional LSTM	1.18	0.91	0.97	1st	Deep Learning	Train/Val
Gradient Boosting	1.23	0.95	0.96	2nd	Ensemble ML	5-fold
Random Forest	1.45	1.12	0.94	3rd	Ensemble ML	5-fold
SARIMA	2.41	1.87	0.82	4th	Statistical	Train/Test

Table 3.2: Final Model Performance Summary (green row = best model)

CHAPTER 4: CONCLUSION

4.1 Summary of Key Findings

This project successfully achieved all of its stated objectives. A complete, end-to-end machine learning pipeline was built for climate trend analysis and temperature forecasting using four years of daily weather data from New Delhi, India. The following key findings emerged from the analysis:

Finding 1: Strong and Stable Seasonality

New Delhi experiences a clear four-season climate cycle with peak mean temperatures of $\sim 35^{\circ}\text{C}$ in May–June and minimum temperatures of $\sim 10^{\circ}\text{C}$ in December–January. This seasonality accounts for over 70% of the total variance in daily temperature and is the dominant signal exploited by all forecasting models.

Finding 2: Statistically Significant Warming Trend

OLS regression on yearly averages revealed a warming rate of $+0.33^{\circ}\text{C}$ per year ($p = 0.008$, $R^2 = 0.94$). This is statistically significant and consistent with regional climate change and urban heat island effects. Extrapolated over 10 years, this implies a $+3.3^{\circ}\text{C}$ increase — a critical planning horizon for city authorities.

Finding 3: Five Major Anomaly Events Detected

Z-score anomaly detection (threshold = 2.5) identified 5 extreme weather events — 2 cold waves (January–February) and 3 heat surges (April–May). These events deviated $7\text{--}9^{\circ}\text{C}$ from their rolling baseline and align with meteorological records for the Delhi region during the study period.

Finding 4: Lag Features are the Most Predictive

Feature importance analysis showed that `temp_lag_1` (31.2%), `temp_7d_avg` (22.1%), and `day_of_year` (14.3%) together explain 67.6% of the model's predictive power. This confirms strong temporal autocorrelation and validates the feature engineering strategy employed in this project.

Finding 5: Bidirectional LSTM Outperforms All Other Models

LSTM achieved $\text{RMSE} = 1.18^{\circ}\text{C}$ and $R^2 = 0.97$, outperforming SARIMA ($\text{RMSE} = 2.41$), Gradient Boosting

(RMSE = 1.23), and Random Forest (RMSE = 1.45). The LSTM's ability to model complex sequential dependencies in 60-day windows gives it a fundamental advantage over tabular and statistical models.

4.2 Practical Applications and Impact

The results of this project have direct and meaningful real-world applications across multiple domains:

- **Agriculture:** Farmers in the Delhi-NCR region can use temperature forecasts to optimize crop sowing dates, irrigation scheduling, and harvest timing. A 90-day forecast with less than 1.5°C average error is sufficient for most agricultural planning decisions.
- **Energy Management:** Delhi's power utilities can use temperature forecasts to anticipate peaks in air conditioning demand during heat waves and heating demand during cold waves, improving grid stability and reducing blackout risk.
- **Public Health:** Extreme heat events (heat waves above 40°C) are associated with increased mortality, especially among elderly populations. Early identification of anomalous temperature events can trigger public health advisories and cooling center activations.
- **Urban Infrastructure:** The detected +0.33°C/year warming trend should inform the design specifications of roads, buildings, and water infrastructure. Materials and designs rated for today's temperature extremes may be inadequate by 2035.
- **Water Resource Management:** Temperature is a key driver of evaporation rates and snowmelt in the Himalayan catchments that feed Delhi's water supply. Accurate temperature forecasting improves water allocation planning.
- **Tourism and Event Planning:** Outdoor events, sports tournaments, and tourism campaigns in the Delhi region can be better planned using reliable multi-month temperature forecasts.

4.3 Comparison with Existing Work

This project's results compare favorably with published work on Delhi climate forecasting. Kharwal (2022) applied a basic Facebook Prophet model to the same dataset but did not report quantitative metrics or compare it against alternative models. This project goes significantly further by: (a) applying four models and rigorously comparing them; (b) engineering 13 domain-informed features; (c) using time-series cross-validation; (d) performing formal anomaly detection and trend analysis; and (e) generating both short-term (90-day) and medium-term (12-month) forecasts.

The LSTM's R^2 of 0.97 is competitive with state-of-the-art results reported in the literature for similar daily temperature forecasting tasks at single-station locations.

4.4 Limitations

Despite the strong results, several limitations should be acknowledged:

- **Short Time Horizon:** The dataset spans only 4 years (2013–2017). While the results are statistically significant, a longer dataset spanning 20–30 years would provide a more robust foundation for detecting climate change signals and training models.
 - **Single Location:** The models are trained on data from a single meteorological station and are specific to New Delhi. They would need to be retrained with location-specific data before being applied elsewhere.
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- **Missing Climate Drivers:** The dataset does not include important climate drivers such as sea surface temperature (El Niño/La Niña indices), monsoon onset date, cloud cover, or solar radiation. Including these could improve model accuracy, particularly for extreme events.
- **LSTM Inference Rollout:** The 90-day autoregressive forecast using Gradient Boosting accumulates errors over time because each prediction is used as input for the next. This error propagation limits the reliability of forecasts beyond ~30 days.
- **No Real-time Deployment:** The current implementation is a research pipeline rather than a production system. A real-time deployment would require a live weather data feed and automated model retraining.

4.5 Future Work

Based on the findings and limitations of this project, the following directions for future work are recommended:

1. **Extended Dataset:** Incorporate post-2020 climate data to include the effects of the COVID-19 lockdown (which temporarily reduced Delhi's air pollution and may have impacted temperatures) and more recent climate events.
2. **Multi-location Forecasting:** Expand the pipeline to simultaneously forecast temperatures for multiple stations across India, enabling spatial interpolation and regional climate maps.
3. **Transformer Models:** Experiment with Transformer-based sequence models (e.g., Temporal Fusion Transformer) which have shown state-of-the-art results on time-series forecasting benchmarks.
4. **External Climate Drivers:** Incorporate ENSO indices, monsoon onset dates, and aerosol optical depth as additional features to improve prediction of extreme events.
5. **Real-time Dashboard:** Deploy the Gradient Boosting model as a real-time web application using Flask or Streamlit, connected to OpenWeatherMap API for live data ingestion.
6. **Probabilistic Forecasting:** Extend the LSTM to output prediction intervals (using Monte Carlo Dropout or Conformal Prediction) to provide uncertainty estimates alongside point forecasts.

4.6 Final Remarks

This project demonstrates that machine learning — and deep learning in particular — can deliver highly accurate, interpretable, and practically valuable climate forecasts. The combination of rigorous preprocessing, thoughtful feature engineering, multiple model architectures, and proper time-series cross-validation resulted in a Bidirectional LSTM achieving $R^2 = 0.97$ on unseen validation data, a level of accuracy that is directly applicable to real-world planning and decision-making.

Beyond the technical results, this project highlights an important truth: the value of a machine learning model lies not just in its architecture, but in the quality of the data preparation, the intelligence of the feature engineering, and the rigor of the evaluation methodology. This project applied best practices at every stage, making its results trustworthy and its conclusions meaningful.

As climate change continues to intensify weather extremes around the world, tools like the ones developed in this project will become increasingly essential. Machine learning offers a scalable, data-driven complement to traditional meteorological methods, and with continued development, has the potential to significantly improve our ability to anticipate and respond to the climate challenges of the coming decades.
